

PROFESSIONAL SUMMARY

AI infrastructure engineer with nearly five years of cloud-native AI/HPC experience, focused since 2024 on production LLM serving. Owns the platform layer between inference engines and application traffic, with end-to-end experience spanning a Kubernetes-native control plane, SGLang runtime, Rust KV-aware Router, cluster-local P/D capacity planning, and offline batch inference. Builds service validation and optimization loops around request success rate, TTFT, TPOT, throughput, queue depth, and ready capacity. Strong in Go and Python, with Rust engineering experience.

EXPERIENCE

Zhejiang Lab

Aug 2021 - Present

Senior Research Specialist · High-Performance Computing Facility Research Center

2024 - Present

LLM Serving / Model-as-a-Service (MaaS) Platform

- Contributed to the architecture and model-service CRDs of a Kubernetes-native MaaS control plane, expressing models, resource profiles, and service forms through one declaration for template-based one-click deployment; implemented inference-instance pools, status and endpoint aggregation, and supported production releases and diagnosis.
- Owned the online serving path for heterogeneous accelerator platforms, connecting the SGLang runtime, KV-aware routing, and cluster-local capacity planning into a closed loop from instance-state awareness and readiness to request placement and replica adjustment.
- Independently designed and implemented a production offline batch-inference platform for data distillation, generation, and evaluation; contributed unified-gateway research and integration, model onboarding, configuration validation, and cross-component production diagnosis.

Engineering Specialist · Intelligent Supercomputing Research Center

Aug 2021 - 2023

Cloud-Native AI / HPC Platform

- Customized Kubernetes Device Plugin and resource-management paths for heterogeneous accelerator discovery, node resource reporting, container-runtime validation, device-level allocation, and fault isolation.
- Adapted PyTorch workloads, models, and distributed job environments across enterprise Linux distributions and heterogeneous accelerator stacks, establishing the platform foundation for later serving and scheduling work.

SELECTED PRODUCTION SYSTEMS

Model-Serving Control Plane and Distributed Inference Runtime

Declarative Serving · Runtime · Readiness

- Contributed to unified model-service CRDs and controllers that express models, resource profiles, service forms, and access endpoints as declarative templates, enabling a one-click path from service declaration and resource orchestration to status aggregation and endpoint publication.
- Owned the SGLang engine adapter, inference-instance registration and discovery, and model publication, unifying topology coordination, layered readiness, and lifecycle management while isolating engine internals from the platform.
- Built layered availability gates and graceful lifecycle handling: scaled-out instances receive traffic only after engine readiness and registration; scale-in drains new traffic, waits for in-flight requests, and then unregisters endpoints, while failed instances are isolated.

KV-aware Router and KV Event Pipeline

Rust Routing · KV Event · Prefix Index

- Implemented a Rust KV-aware Router that combines prefix KV overlap, live load, queue state, and health for request-level instance selection within a model-and-accelerator service domain; integrated SGLang HTTP and gRPC request paths.
- Connected the KV Event adapter, ZMQ-to-NATS bridge, Router event consumption, and prefix index, turning engine KV-cache state into a live routing index and closing the data loop from block events to request placement.

SELECTED PRODUCTION SYSTEMS (CONTINUED)

Cluster-Local Load-Aware P/D Planner	Capacity Modeling · Replica Autoscaling · Observability
• Built a cluster-local load-aware Planner that aggregates Router, inference-instance, and Prometheus signals, combining request load, queue depth, KV state, and offline capacity profiles to continuously compute service replica targets. • Plans prefill and decode replicas independently for P/D-disaggregated services and a unified replica target for co-located services; the Planner owns capacity decisions while the control plane and Kubernetes create and place resources. • Closed the production loop from load sensing and capacity decisions to elastic scaling and outcome readback; validated scale-out under rising load and scale-in after load subsided using request success rate, TTFT/TPOT, queue depth, and ready capacity.	
Multi-Tenant Offline Batch-Inference Platform	OpenAI Batch API · Scheduling · Recovery
• Built an OpenAI Batch API-compatible platform for large, long-running data-distillation, generation, and evaluation workloads; JSONL submissions are streamed, sharded, scheduled concurrently, and aggregated automatically. • Persists task progress and uses Kubernetes Lease leadership plus shard reassignment so schedulers or executors can fail without restarting an entire batch. • Uses at-least-once backend execution with result-level de-duplication and custom_id-based skipping, providing request-level error isolation and stable long-running operation.	

CORE TECHNOLOGIES

Runtime & Data Plane: distributed inference runtime, SGLang, KV Event Plane, ZMQ/NATS, P/D disaggregation
Control Plane & Scheduling: Go, Kubernetes Operator, CRD/controller, inference-instance pools, capacity modeling, P/D replica autoscaling, Device Plugin, Helm, ArgoCD
Routing & Gateway: Rust, KV-aware routing, HTTP/gRPC routing, prefix index, LiteLLM research and integration, OpenAI/Anthropic APIs, authentication and rate-limit configuration
Batch & Observability: Python, OpenAI Batch API, Redis, MySQL, Kubernetes Lease, Prometheus, QPS/TTFT/TPOT, request tracing

EDUCATION

Arizona State University - M.S. in Computer Science, GPA 3.97 / 4.00, May 2019 - Dec 2020
Arizona State University - B.S. in Computer Science, Aug 2015 - May 2019

PUBLICATION & INTELLECTUAL PROPERTY

Prompt2Length: A Lightweight Model for Predicting Output Length in LLM Inference - third author, IEEE AIAHPC 2025; co-inventor on two granted invention patents and named contributor on two registered software copyrights.

LANGUAGES

Mandarin Chinese (native); English (full professional proficiency).
